

GapMind Prediction

Experimental Outcome

Growth (+)

No Growth (–)

Pathway Present (+)

Pathway Absent (–)

CONCORDANT

True Positive

- ✓ Genes present & functional
- ✓ Pathway correctly annotated
- ✓ Experiment accurately measured

High-quality training samples

DISCORDANT

False Negative

Possible Causes:

- Annotation: homolog not detected
- Biology: alternative pathway
- Biology: promiscuous enzymes
- Measurement: false positive
- Media: growth on base media

Growth via unknown mechanism

DISCORDANT

False Positive

Possible Causes:

- Regulation: pathway repressed
- Measurement: insufficient time
- Media: missing cofactors
- Annotation: false positive call
- Biology: incomplete pathway

Genes present but not functional

CONCORDANT

True Negative

- ✓ Genes absent
- ✓ No alternative pathway
- ✓ Experiment accurately measured

High-quality training samples

■ Concordant (use for training)

■ Discordant (analyze for data quality insights)